MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : OECAID3701A/OECAIML701A Internet of Things

UPID : 007816

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

   (I) Why is monitoring air quality important for IoT applications?

   (II) What is the basic working principle of a resistive sensor?

   (III) Define constant phase impedance.

   (IV) What is the primary function of a sensing element in a smart sensor?

   (V) What is the purpose of interfacing electronic circuits with smart sensors?

   (VI) What does the sensitivity of a sensor refer to?

   (VII) What is the function of a pressure sensor?

   (VIII) What is one advantage of using integrated circuits (ICs) in smart sensor designs?

   (IX) How is the equivalent circuit of a sensor used in modeling?

   (X) Explain the purpose of a power management system in smart sensors.

   (XI) What is a common application of smart sensors in industrial automation?

   (XII) Give an example of a smart sensor application in agriculture.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Explain the role of smart sensors in enhancing daily life through health monitoring. [5]
3. Discuss the effects of adverse environmental parameters on human health and how IoT can mitigate them. [5]
4. Describe the structure and function of a resistive humidity sensor. [5]
5. Explain the architecture of smart sensors, detailing the key components and their respective roles in sensor functionality https://www.makaut.com [5]
6. How can adverse environmental conditions affect the performance of IoT networks, and what are the solutions? [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. Evaluate the significance of real-time environmental monitoring in healthcare and urban planning using IoT technologies. [15]
8. Discuss in detail the working principles of capacitive, resistive, and SAW sensors. Provide a practical example for each sensor type. [15]
9. (a) Describe the role of impedance spectroscopy in the characterization of sensors. [5]

   (b) Include a discussion of its methodology and applications in various sensing environments. [10]
10. (a) Evaluate the fabrication techniques of electrodes in smart sensors, focusing on screen printing, photolithography, and electroplating. [5]

    (b) Discuss their processes, advantages, limitations, and applications. [10]
11. Illustrate the future scope of research in smart sensors, focusing on emerging technologies and their potential applications across various sectors. [15]

*** END OF PAPER ***

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